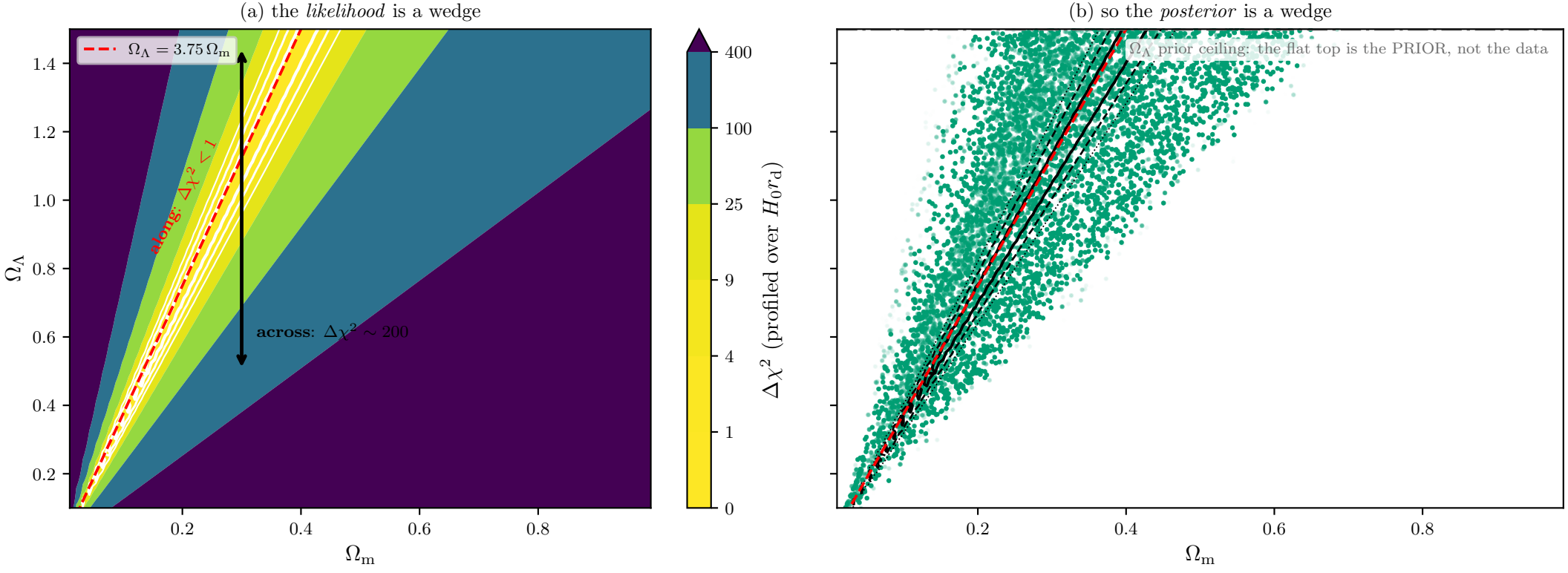
Why the $(\Omega_\mathrm{m}, \Omega_\Lambda)$ posterior is a wedge pointing at the origin



Scaling *both* Ω's together rescales H by one common factor at every epoch, which $E(z) = H/H(a{=}1)$ divides out — so the fit is flat along $\Omega_\Lambda \propto \Omega_\mathrm{m}$ ($\Delta\chi^2 < 1$ over a factor 7.5 in Ω_m).
Changing *one* of them moves $\Omega_\Lambda/\Omega_\mathrm{m}$, hence the epoch when Λ overtakes matter ($z_\Lambda \approx 0.54$, right where DESI observes) — so the fit collapses across the wedge ($\Delta\chi^2 \sim 200$).
The valley is broken only by $\Omega_r a^{-4}$, which is $< 0.1\%$ of the source at $z < 2.6$. The wedge therefore runs until the *prior* stops it, not the data.